

CARMA-Based Deseasonalization, Calibration and Pricing in Renewable Energy Markets

August 17, 2026

Abstract

1 Introduction

This paper builds a compact empirical framework for pricing renewable-energy contracts whose payoffs depend jointly on German day-ahead electricity prices and weather-driven production factors. The workflow first removes deterministic seasonality from prices, temperature, solar capacity factors, and wind capacity factors. It then fits marginal CARMA models to the resulting stochastic factors, recovers their driving noises, and estimates one-way shock couplings from temperature, solar, and wind into the price driver. The calibrated system is then ready for joint simulation and pricing.

2 Data and Deseasonalization

2.1 Common Setup

Each raw series is first mapped to a coordinate in which an additive model is appropriate. Denote this transformed observable by Y_t^F , where $F \in \{P, T, Q, W\}$ refers to price, temperature, solar, and wind. We write

$$Y_t^F = \Lambda_t^F + X_t^F. \quad (1)$$

Here Λ_t^F is a deterministic seasonal component and X_t^F is the stochastic residual used in the CARMA calibration. This section specifies the coordinate Y_t^F , the seasonal structure used for Λ_t^F , and the diagnostic evidence left after deseasonalization. The remaining serial dependence is then modeled stochastically.

For the physical factors T , Q , and W , the seasonal component is a linear regression on calendar and Fourier covariates after the relevant transformation has been applied, with coefficients estimated by OLS. Prices use a separate normalized seasonal construction, described below, because the intraday price profile is more irregular.

2.2 Electricity Prices

We use hourly German day-ahead prices for the DE–LU bidding zone. The CARMA calibration sample is 2023–2025 in UTC, while fig. 1 also shows 2021–2022. As seen in the figure, the 2023–2025 window is used to exclude the exceptional price spikes associated with the energy crisis and the war in Ukraine.

Since the spot price S_t can be negative, the stochastic price factor is defined in shifted log coordinates,

$$\tilde{S}_t = \log(S_t + \delta), \quad \delta = 1000 \text{ EUR/MWh}, \quad S_t = \exp(\tilde{S}_t) - \delta. \quad (2)$$

Thus $Y_t^P = \tilde{S}_t$. The price seasonality follows the electricity-price deseasonalization approach discussed in [1]. We estimate Λ_t^P with a multiplicative seasonal weight in log-price coordinates,

$$\hat{\Lambda}_t^P = \bar{p} \hat{F}_{d(t)}^Y \hat{F}_t^D. \quad (3)$$

The daily factor $\hat{F}^Y$ is fitted from normalized daily mean log-prices using weekday and month effects, with August split into early and late subperiods. The intraday factor $\hat{F}^D$ is fitted from hour-of-day profiles within 20 calendar classes, then normalized so that it only captures shape. The estimated residual is the price factor used in the CARMA calibration.

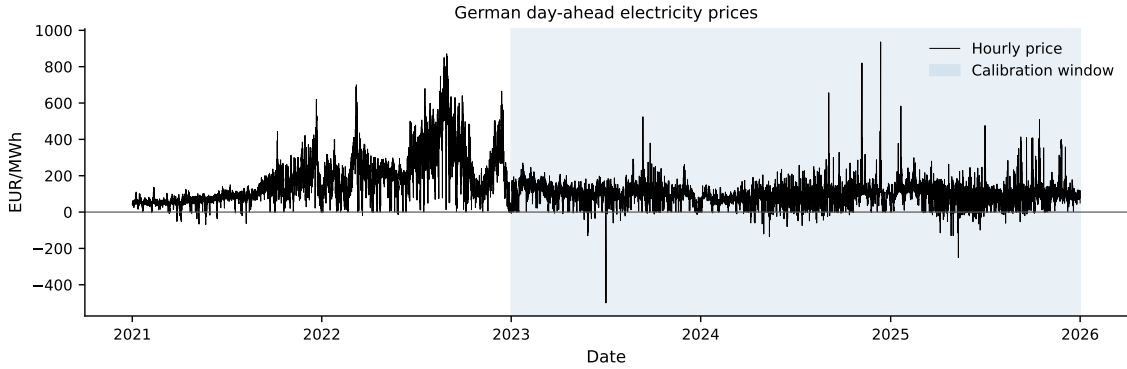


Figure 1: Hourly German day-ahead electricity prices from 2021 to 2025. The shaded region is the 2023–2025 calibration window.

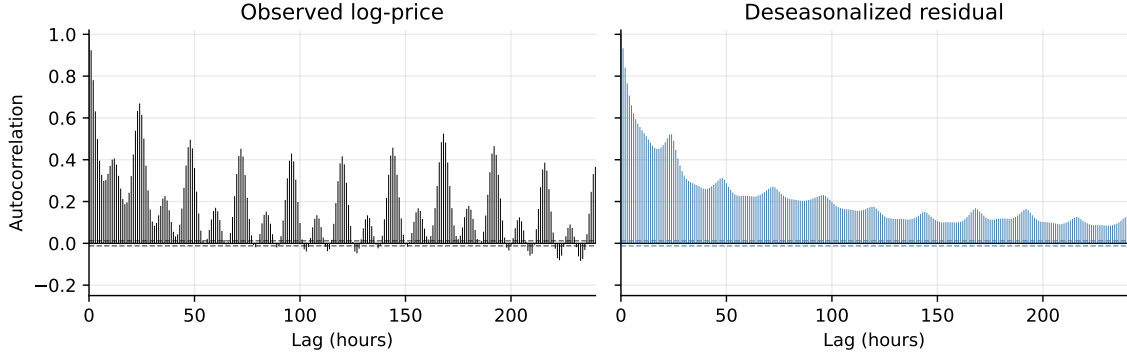


Figure 2: Empirical autocorrelation of the shifted log-price and of the deseasonalized price residual over 240 hourly lags.

2.3 Temperature

We use hourly 2-meter Berlin temperature over the same 2023–2025 UTC sample. No transform is needed, so $Y_t^T = T_t$. The seasonal regression uses a compact Fourier basis, where τ_t denotes the elapsed number of hours and $P_y = 365.25 \times 24$.

$$\Lambda_t^T = \beta_0 + \beta_1 \tau_t + \sum_{k=1}^3 \left[a_k(t) \cos\left(\frac{2\pi k \tau_t}{24}\right) + b_k(t) \sin\left(\frac{2\pi k \tau_t}{24}\right) \right] + a_y \cos\left(\frac{2\pi \tau_t}{P_y}\right) + b_y \sin\left(\frac{2\pi \tau_t}{P_y}\right). \quad (4)$$

The daily-cycle coefficients $a_k(t)$ and $b_k(t)$ are annual Fourier functions, so the intraday temperature profile can vary smoothly over the year. Figure 3 shows the residual dependence left for the CARMA model.

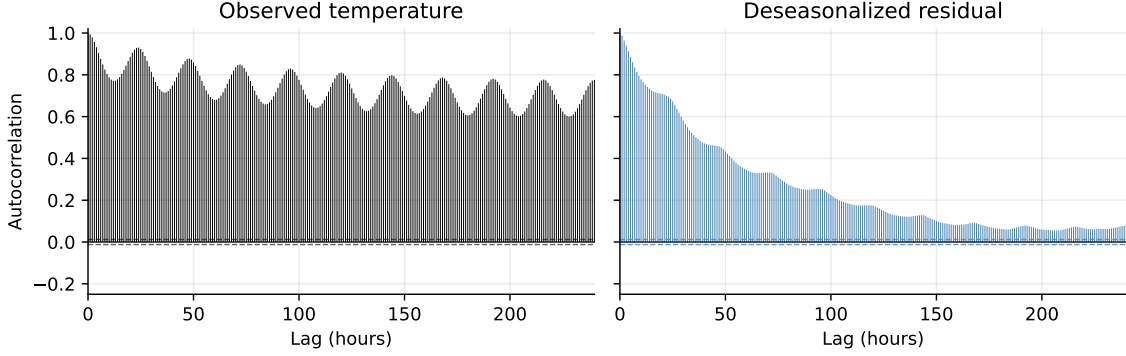


Figure 3: Empirical autocorrelation of observed temperature and of the deseasonalized temperature residual over 240 hourly lags.

2.4 Solar Capacity Factors

Let $Q_t \in [0, 1]$ be the German solar capacity factor. Solar cannot be treated as a simple additive bounded process because production is structurally zero at night and the daylight support changes over the year. We first estimate an empirical clear-sky envelope $\hat{C}_t = \hat{C}_{d(t), h(t)}$ from the 98% quantile of Q_t by day-of-year and hour-of-day cells, with missing-cell completion and circular 21-day smoothing.

The stochastic solar variable is defined as a clear-sky shortfall,

$$R_t = \text{clip}_{[0,1]} \left(1 - \frac{Q_t}{\max(\hat{C}_t, 10^{-4})} \right), \quad Z_t = \text{clip}_{[10^{-4}, 1-10^{-4}]} \left(\frac{R_t - 10^{-4}}{0.9998} \right), \quad (5)$$

$$Y_t^Q = \log \left(\frac{Z_t}{1 - Z_t} \right).$$

A positive solar latent shock therefore means larger cloudiness or solar shortfall, not larger physical production. Deseasonalization is performed with Y_t^Q as the transformed coordinate. The seasonal regression combines hour and month effects, annual Fourier terms, and hour-by-annual interactions. We write $H_h(t) = \mathbf{1}_{\{h(t)=h\}}$, $M_m(t) = \mathbf{1}_{\{\text{month}(t)=m\}}$, and $\phi_t = 2\pi d(t)/365.25$. The regression is

$$\Lambda_t^Q = \gamma_0 + \sum_{h=1}^{23} \gamma_h H_h(t) + \sum_{m=2}^{12} \eta_m M_m(t) + \sum_{k=1}^3 [a_k \cos(k\phi_t) + b_k \sin(k\phi_t)]$$

$$+ \sum_{h=1}^{23} H_h(t) \sum_{k=1}^2 [u_{h,k} \cos(k\phi_t) + v_{h,k} \sin(k\phi_t)]. \quad (6)$$

For physical diagnostics, the fitted latent seasonality is mapped back through the inverse clear-sky transform.

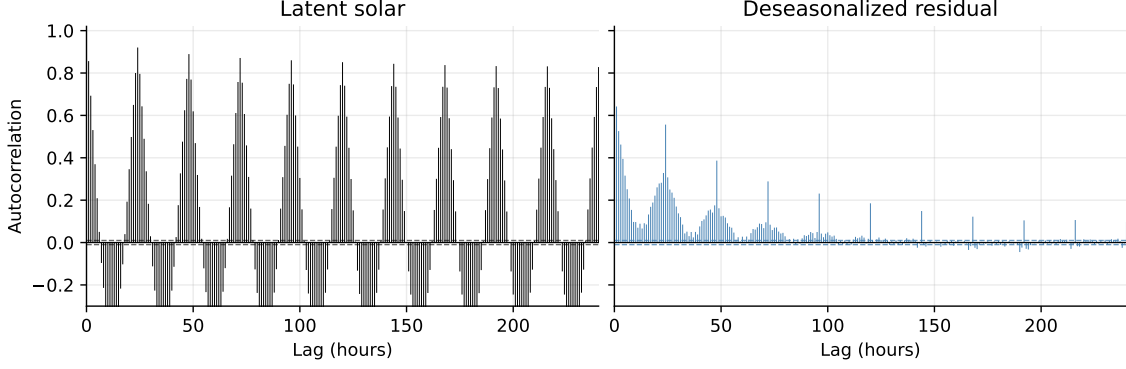


Figure 4: Empirical autocorrelation of the latent solar coordinate and of the deseasonalized solar residual over 240 hourly lags.

2.5 Wind Capacity Factors

Let $W_t \in [0, 1]$ be the German wind capacity factor, aligned with the price calendar in UTC. Wind has no structural day-night support, but it is bounded and persistent. We therefore work in logit coordinates,

$$W_t^\varepsilon = \text{clip}_{[10^{-4}, 1-10^{-4}]}(W_t), \quad Y_t^W = \log\left(\frac{W_t^\varepsilon}{1 - W_t^\varepsilon}\right), \quad W_t = \sigma(Y_t^W). \quad (7)$$

The transform is monotone, so a positive wind latent shock corresponds to higher-than-seasonal physical wind production. The seasonal regression includes hour, weekday, month, and annual Fourier effects. Physical-scale diagnostics are obtained by mapping $\hat{\Lambda}_t^W$ back through the logistic function.

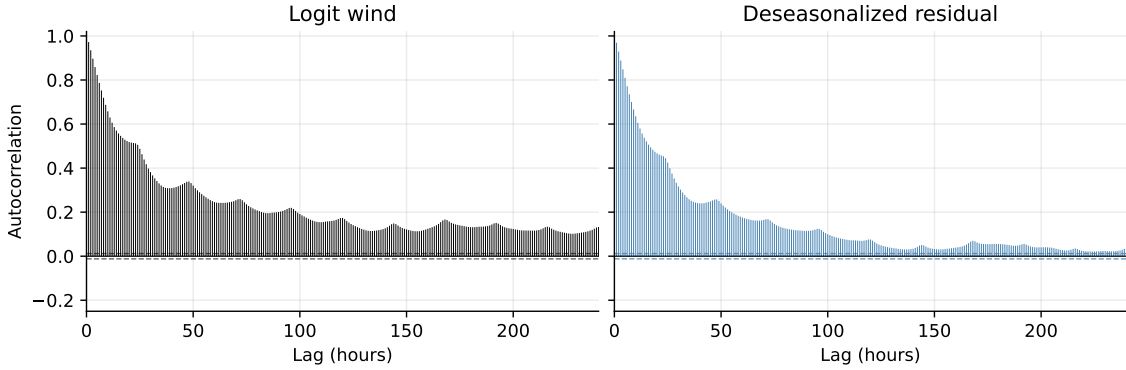


Figure 5: Empirical autocorrelation of the logit wind capacity factor and of the deseasonalized wind residual over 240 hourly lags.

3 Marginal CARMA Calibration

3.1 Calibration Framework

Each deseasonalized marginal factor is modeled as a stationary Levy-driven CARMA process in state-space form [2]. For a CARMA(p, q),

$$dG_t = AG_t dt + e_p dL_t, \quad X_t = b^\top G_t, \quad (8)$$

where L is a one-dimensional Levy driver and A is the companion matrix of

$$a(z) = z^p + a_1 z^{p-1} + \dots + a_p. \quad (9)$$

The roots of a are required to have negative real parts. The moving-average vector is written as $b = (b_0, \dots, b_q, 0, \dots, 0)^\top$, with the last non-zero coefficient normalized to one. Throughout the paper we use the full moving-average convention $q = p - 1$, unless stated otherwise.

The order is initialized from the empirical autocorrelation function. A real root contributes a monotone exponential decay, while a complex conjugate pair contributes a damped oscillation. We therefore fit

$$\rho_{\text{ms}}(h) = \sum_{r=1}^R w_r e^{-\kappa_r h} + \sum_{\ell=1}^C w_{o,\ell} e^{-\kappa_{o,\ell} h} \cos(\omega_\ell h), \quad \sum_r w_r + \sum_\ell w_{o,\ell} = 1, \quad (10)$$

with non-negative weights. In the applications below there is one daily oscillatory pair, with $\omega = 2\pi/24$, and the fitted ACF is spectrally factorized to obtain a stable initial CARMA model.

Starting from this initialization, the final marginal parameters are estimated by exact Gaussian prediction-error QMLE, following the Kalman likelihood construction for sampled CARMA processes [3]. For given CARMA coefficients, the Kalman recursion gives one-step prediction errors e_i and innovation variances r_i under unit Levy variance. Since the Levy drift enters as a constant mean shift, the adjusted innovation is

$$\varepsilon_i(m) = e_i - m c_i.$$

The drift rate and variance rate are profiled as

$$\hat{m} = \frac{\sum_i e_i c_i / r_i}{\sum_i c_i^2 / r_i}, \quad \hat{\nu}^2 = \frac{1}{n} \sum_i \frac{\varepsilon_i(\hat{m})^2}{r_i}. \quad (11)$$

The optimized criterion is the reduced Gaussian prediction-error likelihood, augmented when needed by a soft ACF penalty,

$$\mathcal{J}(\theta) = \log(\hat{\nu}^2) + \frac{1}{n} \sum_i \log r_i + \lambda_{\text{acf}} \frac{1}{|H|} \sum_{h \in H} \omega_h (\rho_\theta(h) - \hat{\rho}(h))^2. \quad (12)$$

The penalty stabilizes long-lag dependence without replacing the likelihood criterion. Its value is reported separately for each marginal.

Once the CARMA parameters are fixed, the hourly Levy increments are recovered from the smoothed state path, as in the QMLE framework of Brockwell and Lindner [3]. Let $\hat{G}_t$ denote the RTS-smoothed state, including the profiled drift correction $\hat{m}(-A)^{-1}e_p$. Choose a real root λ_r of a , taken in practice as the slowest real root, and let E be the matrix of companion eigenvectors. The corresponding modal coordinate satisfies a scalar driven equation, giving

$$\hat{Z}_r(t) = b(\lambda_r) e_r^\top E^{-1} \hat{G}_t, \quad d\hat{Z}_r(t) = \lambda_r \hat{Z}_r(t) dt + \alpha_r dL_t, \quad \alpha_r = \frac{b(\lambda_r)}{a'(\lambda_r)}, \quad (13)$$

and therefore

$$\hat{L}_t - \hat{L}_0 = \frac{\hat{Z}_r(t) - \hat{Z}_r(0) - \lambda_r \int_0^t \hat{Z}_r(s) ds}{\alpha_r}. \quad (14)$$

On the hourly grid the integral is evaluated numerically, and the empirical driver sample used below is $\Delta \hat{L}_i = \hat{L}_{t_i} - \hat{L}_{t_{i-1}}$.

3.2 German Spot Prices

The deseasonalized German log-price factor $\hat{X}^P$ is calibrated with the procedure of section 3.1. To represent a decaying ACF with damped 24-hour oscillations, the initialization combines one daily oscillatory pair with three real half-lives. The first real scale is expected to be

hourly, the second daily, and the third is fixed at 120 d during initialization to impose a slow component for option-pricing horizons.

$$\rho_{\text{ms}}(h) = \sum_{i=1}^3 w_i \exp(-\kappa_i h) + w_4 \exp(-\kappa_4 h) \cos\left(\frac{2\pi h}{24}\right), \quad \sum_{i=1}^4 w_i = 1, \quad w_i \geq 0, \quad (15)$$

Here h is measured in hours, and the oscillatory frequency is fixed at $2\pi/24$. The additional long real half-life is specific to the price model. It is introduced only at initialization to keep slow price persistence available for pricing, while the temperature, solar, and wind marginals below do not require such an imposed long mode. The fitted decomposition has three real roots and one complex conjugate pair, which gives the autoregressive order

$$p = 3 + 2 = 5.$$

We use the full moving-average order $q = p - 1 = 4$. Spectral factorization of the fitted ACF then gives the initial CARMA(5,4) model.

Table 1: German price CARMA(5,4) half-lives before and after penalized QMLE.

Root	Half-life after initialization	Half-life after QMLE
λ_1	2.63 h	4.52 h
λ_2	1.20 d	1.50 d
λ_o	7.21 d	7.21 d
λ_3	120 d	82.4 d

The QMLE step uses $\lambda_{\text{acf}} = 10$. The daily pair remains fixed, whereas the three real half-lives, including the slow one, and the moving-average coefficients are re-estimated. The final log-likelihood is 77709. As shown in fig. 6, the ACF penalty keeps the final CARMA autocorrelation close to the multiscale initialization while allowing the likelihood to improve.

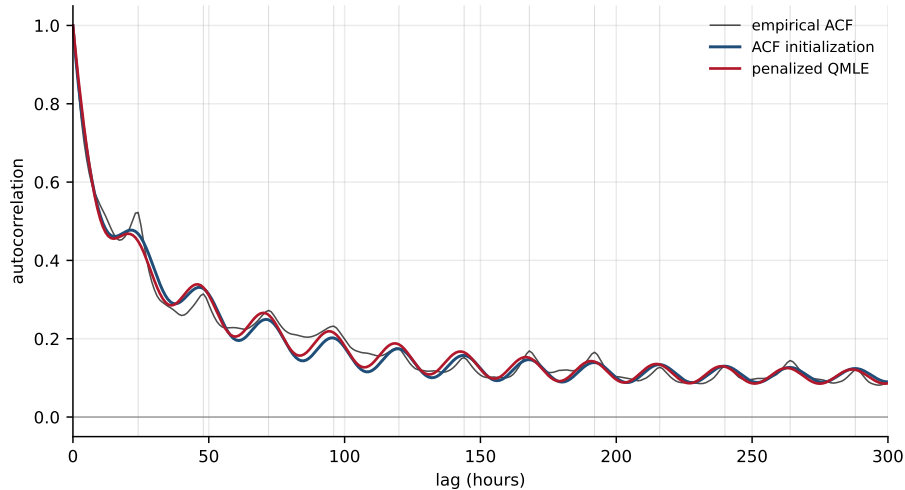


Figure 6: German price ACF calibration over 300 hourly lags, comparing the empirical autocorrelation, the multiscale initialization, and the final CARMA(5,4) autocorrelation after QMLE.

Table 2 shows that the penalized QMLE improves the Gaussian log-likelihood relative to the ACF initialization, although it remains below the unconstrained ARMA(5,4) benchmark. Pure QMLE can increase this likelihood further, as discussed below, but only by shortening the slow price component too much.

Table 2: Gaussian log-likelihood comparison on the same deseasonalized German log-price residuals.

Model	Log-likelihood
CARMA(5,4), ACF initialization	77562
CARMA(5,4), QMLE	77709
ARMA(1,0)	77454
ARMA(5,4)	78661

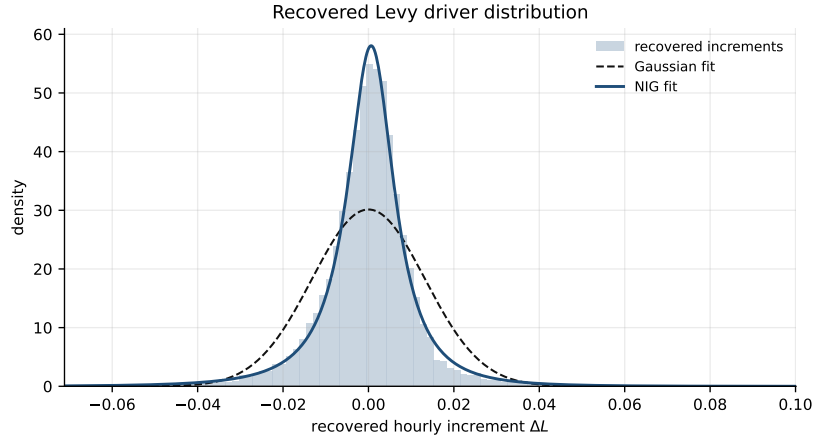


Figure 7: Recovered hourly Levy driver increments for the German price CARMA model, with Gaussian and NIG distribution fits.

After calibration, the driver increments recovered by eq. (14) are fitted with a Gaussian law and with a normal inverse Gaussian law. The NIG specification is preferred according to the AIC criterion, consistently with fig. 7. The NIG density better matches the asymmetry and tail thickness of the recovered increments than the Gaussian benchmark.

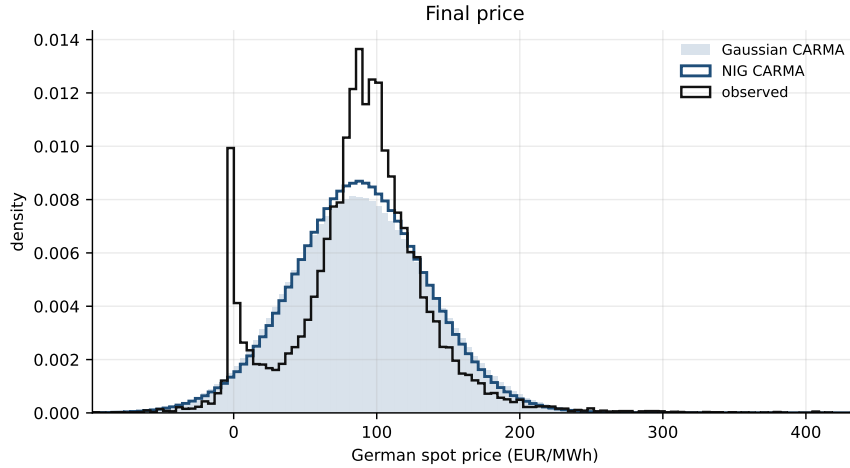


Figure 8: Distribution check for the final German spot price in EUR/MWh, comparing the observed data with prices simulated from the Gaussian and NIG CARMA marginals.

Figure 8 maps the simulated CARMA residuals back to EUR/MWh using the same seasonal component. The NIG-driven model improves the distribution relative to the Gaussian-driven model, but neither specification reproduces the bimodality of the empirical price distribution, in particular the concentration around zero EUR/MWh. In the European day-ahead market, hourly prices are obtained by uniform-price auctions from the intersection of aggregated supply and demand curves. During hours with low residual demand, renewable produc-

tion and other low marginal-cost or must-run units frequently set, or strongly constrain, the clearing price near zero. During tighter hours, the marginal unit is usually thermal or flexible generation with a strictly positive bid. This market-clearing mechanism creates regime-like mass in the final price distribution and cannot be reproduced by a smooth continuous-time CARMA marginal factor alone.

We also ran a pure exact QMLE without the ACF penalty ($\lambda_{\text{acf}} = 0$), using semi-global restarts from the same CARMA(5,4) structure. The best run over 20 restarts increases the Gaussian log-likelihood because the objective only rewards one-step prediction fit. It does not penalize the loss of long-lag ACF structure.

Table 3: Penalized versus pure QMLE for the German price CARMA(5,4).

Quantity	Retained QMLE	Pure QMLE
ACF penalty λ_{acf}	10	0
Log-likelihood	77709	78808
λ_1 half-life	4.52 h	0.510 h
λ_2 half-life	1.50 d	1.20 h
λ_o half-life	7.21 d	5.20 d
λ_3 half-life	82.4 d	1.40 d

The pure QMLE is therefore better as a one-step Gaussian likelihood fit, but it collapses the slow price factor from several months to about one day. This choice is not appropriate for the pricing step, because the model would lose the long-horizon persistence imposed by the ACF initialization and retained by the penalized QMLE.

3.3 German Temperature

The stochastic temperature factor $\hat{X}^T$ is the hourly Berlin temperature residual after deterministic Fourier deseasonalization. It is modeled with the stationary CARMA structure of eq. (8). Its empirical ACF is smoother than the electricity price ACF, but it still contains a daily oscillatory component. The initialization therefore uses two real modes and one daily pair.

$$\rho_{\text{ms}}^T(h) = w_f e^{-\kappa_f h} + w_s e^{-\kappa_s h} + w_d e^{-\kappa_d h} \cos\left(\frac{2\pi h}{24}\right), \quad w_f + w_s + w_d = 1, \quad w_f, w_s, w_d \geq 0. \quad (16)$$

The two real components represent a relatively fast decay and a slower persistent component, while the complex conjugate pair enforces the 24-hour oscillation. The resulting order is

$$p = 2 + 2 = 4, \quad q = p - 1 = 3.$$

The daily frequency is fixed in the ACF step and the daily AR pair is kept fixed in QMLE. The two real AR roots and the moving-average polynomial are re-estimated.

Table 4: German temperature CARMA(4,3) half-lives before and after penalized QMLE.

Root	Half-life after initialization	Half-life after QMLE
λ_1	1.06 d	0.530 h
λ_2	2.42 d	1.59 d
λ_o	3.03 d	3.03 d

For temperature, the QMLE step uses $\lambda_{\text{acf}} = 0.1$. This penalty is lower than for prices because the Gaussian likelihood is on a different numerical scale. The trade-off is visible in

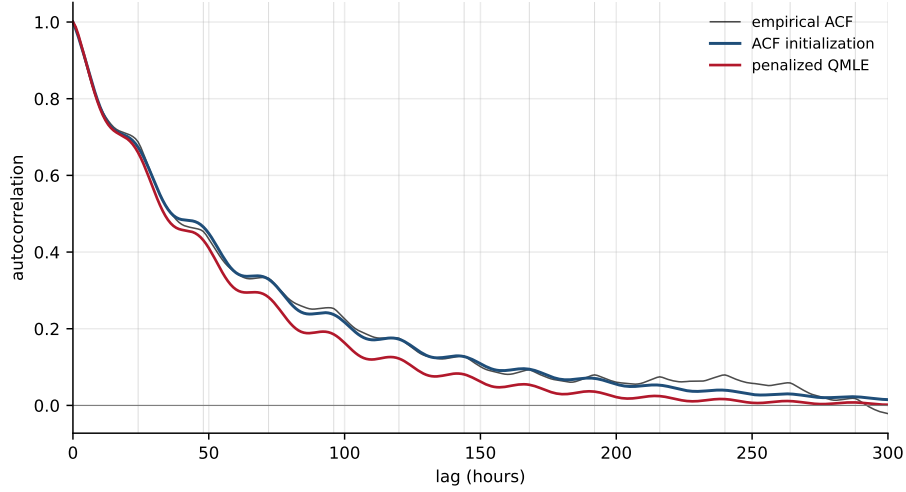


Figure 9: German temperature ACF calibration over 300 hourly lags, comparing the empirical autocorrelation, the multiscale initialization, and the final CARMA(4,3) autocorrelation after QMLE.

fig. 9, where the final ACF is less tightly matched than in the price calibration. The weaker penalty nevertheless improves the likelihood fit relative to the ACF initialization, with a final log-likelihood of -22941 , while remaining below the ARMA(4,3) benchmark.

Table 5: Gaussian log-likelihood comparison on the same deseasonalized German temperature residuals.

Model	Log-likelihood
CARMA(4,3), ACF initialization	-23760
CARMA(4,3), QMLE	-22941
ARMA(1,0)	-24612
ARMA(4,3)	-23173

After QMLE, the slow real half-life is about one and a half days, close to the solar value and much shorter than the wind and price slow modes. Temperature is therefore the cleanest two-scale physical marginal, with a fast adjustment, a one-to-two-day persistence scale, and the fixed daily oscillation.

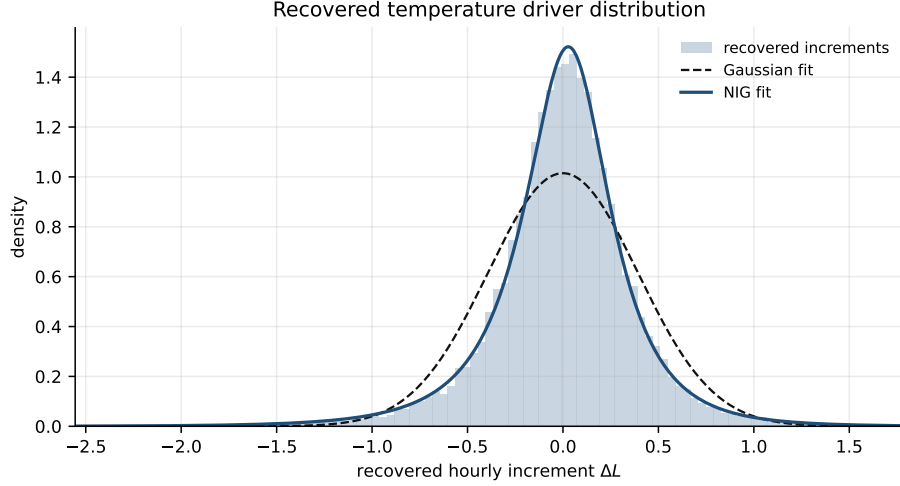


Figure 10: Recovered hourly Levy driver increments for the German temperature CARMA model. The NIG fit captures the asymmetric and heavy-tailed driver distribution better than the Gaussian fit.

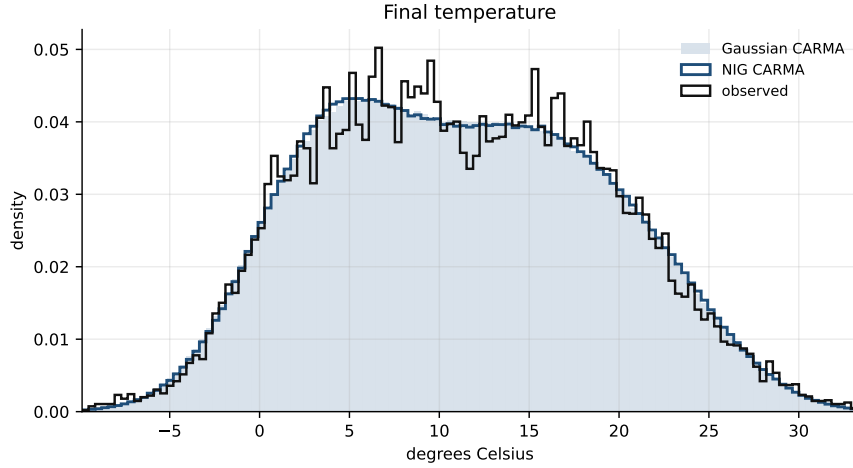


Figure 11: Distribution check for the final German temperature level, comparing the observed data with paths simulated from the Gaussian and NIG CARMA marginals.

3.4 German Solar Production

The solar marginal model is fitted to the deseasonalized latent residual $\hat{X}^Q$ defined after the clear-sky normalization and logit transformation in section 2.4. The physical capacity factor is bounded and has structural nighttime zeros, so the CARMA model is not applied directly to Q_t . It is estimated in the transformed coordinate, and simulated paths are mapped back through the inverse clear-sky transformation.

As for temperature, we use a CARMA(4,3) with two real modes and one daily oscillatory pair. The solar ACF is more specific because it contains periodic peaks every 24 hours, which must be preserved in calibration. The daily pair is fixed after the ACF step, while QMLE re-estimates the two real persistence scales and the moving-average polynomial.

Table 6: German solar CARMA(4,3) half-lives before and after penalized QMLE.

Root	Half-life after initialization	Half-life after QMLE
λ_1	0.506 h	0.512 h
λ_2	1.74 d	1.57 d
λ_o	2.93 d	2.93 d

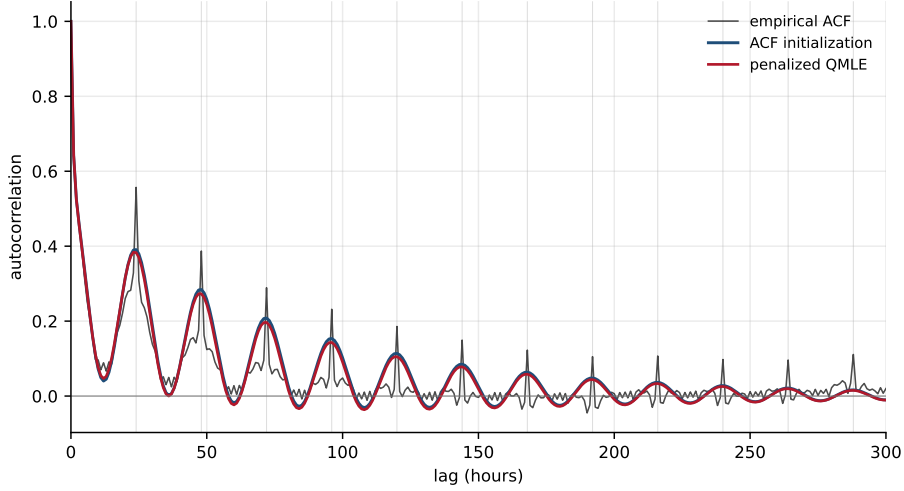


Figure 12: German solar latent-residual ACF calibration over 300 hourly lags, comparing the empirical autocorrelation, the multiscale initialization, and the final CARMA(4,3) autocorrelation after QMLE.

For solar, the QMLE step uses $\lambda_{\text{acf}} = 1000$. This larger penalty is chosen to preserve the latent ACF structure, in particular its daily oscillatory component. This choice is also supported by the likelihood comparison, as the ACF initialization already has a higher log-likelihood than the ARMA(4,3) benchmark. QMLE then refines the fit only slightly, with a final log-likelihood of -84623 . This leads to only small changes in both the estimated half-lives and the log-likelihood.

Table 7: Gaussian log-likelihood comparison on the same deseasonalized German solar latent residuals.

Model	Log-likelihood
CARMA(4,3), ACF initialization	-84637
CARMA(4,3), QMLE	-84623
ARMA(1,0)	-86462
ARMA(4,3)	-84786

The solar CARMA(4,3) has roots close to those obtained for temperature. One real half-life is below one hour, another is around one and a half days, and the daily oscillations have a damping half-life of about three days.

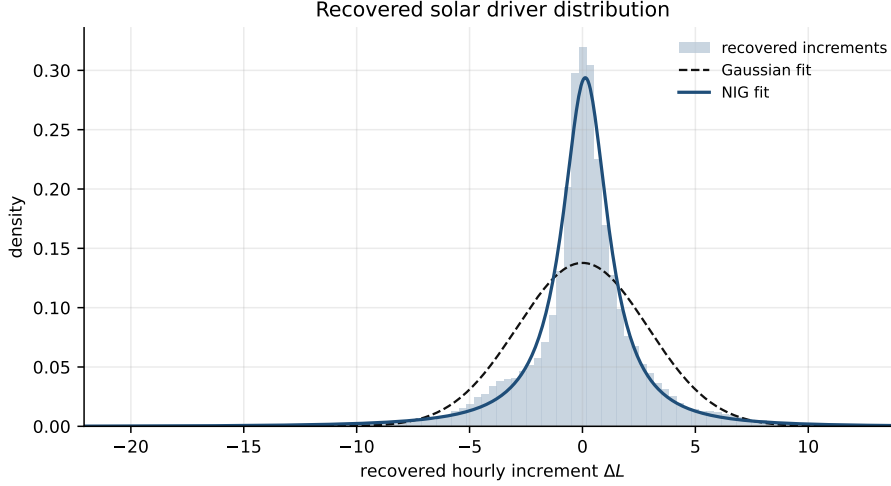


Figure 13: Recovered hourly Levy driver increments for the German solar CARMA model. The NIG fit captures the asymmetric and heavy-tailed driver distribution better than the Gaussian fit.

3.5 German Wind Production

The wind marginal model is fitted to the deseasonalized logit residual $\hat{X}^W$ defined in section 2.5. As for solar, the CARMA model is estimated in an unconstrained latent coordinate and simulated paths are mapped back to capacity factors with the inverse logit transform. This keeps the continuous-time model away from the physical bounds 0 and 1.

As for temperature and solar, we use a CARMA(4,3) with two real modes and one daily oscillatory pair. The wind ACF is less tied to a deterministic day-night structure than solar, but it has longer persistence. The daily pair is fixed after the ACF step, while QMLE re-estimates the two real persistence scales and the moving-average polynomial.

Table 8: German wind CARMA(4,3) half-lives before and after penalized QMLE.

Root	Half-life after initialization	Half-life after QMLE
λ_1	16.1 h	20.9 h
λ_o	3.28 d	3.28 d
λ_2	7.44 d	37.6 d

For wind, the QMLE step uses $\lambda_{acf} = 0.1$, giving a final log-likelihood of -6984 . The fit improves the ACF initialization but remains below the ARMA(4,3) benchmark in log-likelihood.

Table 9: Gaussian log-likelihood comparison on the same deseasonalized German wind logit residuals.

Model	Log-likelihood
CARMA(4,3), ACF initialization	-7013
CARMA(4,3), QMLE	-6984
ARMA(1,0)	-7262
ARMA(4,3)	-6853

The main difference with solar and temperature is the slow real mode, whose half-life increases to about 38 days after QMLE. It represents persistent wind-regime variation in the latent residual.

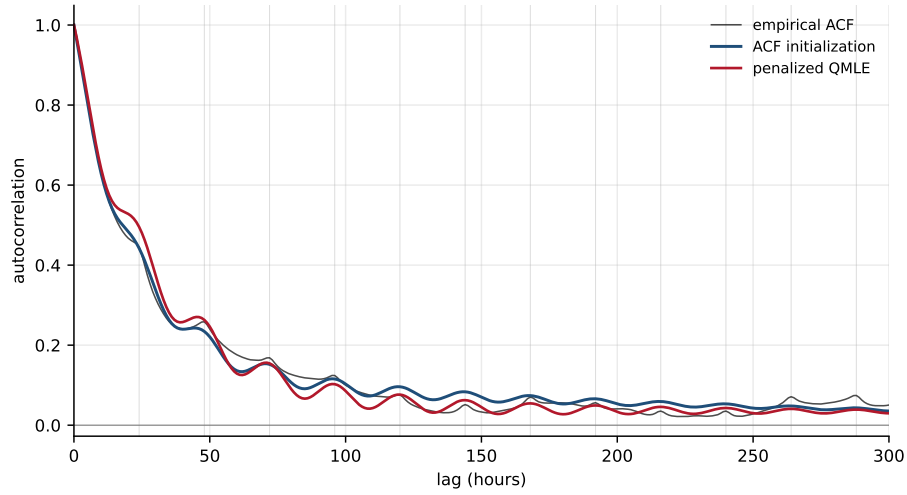


Figure 14: German wind logit-residual ACF calibration over 300 hourly lags, comparing the empirical autocorrelation, the multiscale initialization, and the final CARMA(4,3) autocorrelation after QMLE.

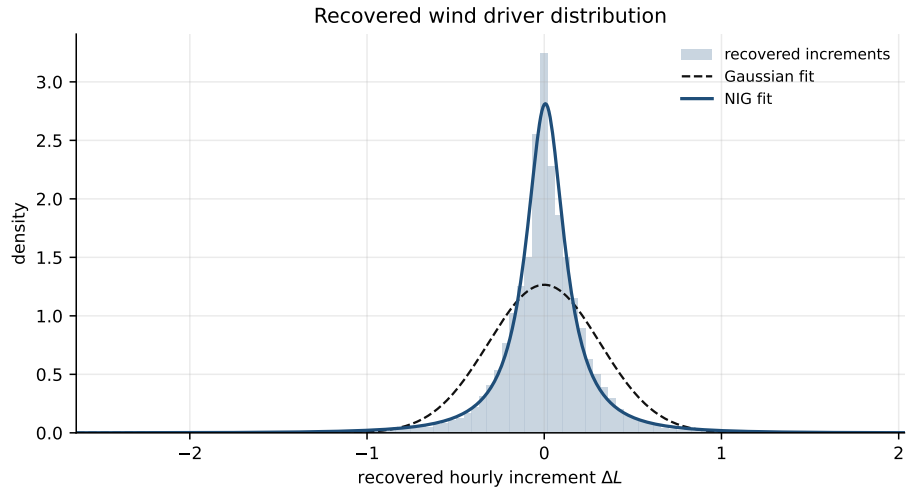


Figure 15: Recovered hourly Levy driver increments for the German wind CARMA model. The NIG fit captures the narrow center and heavy tails better than the Gaussian fit.

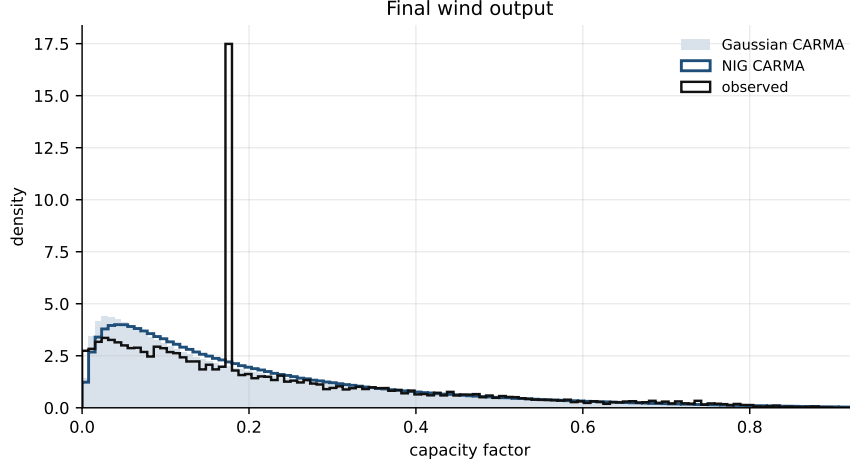


Figure 16: Distribution check for the final German wind capacity factor, comparing the observed data with paths simulated from the Gaussian and NIG CARMA marginals.

4 Coupling Calibration

4.1 Coupling Framework

The marginal CARMA models are kept fixed. For a weather or renewable factor $F \in \{T, S, W\}$, dL^F denotes the calibrated marginal driver, with its variance scale included as in eq. (8). The coupled price model transmits this factor driver into the price driver.

$$\begin{aligned} dZ_t^F &= A_F Z_t^F dt + e_F dL_t^F, & Y_t^F &= b_F^\top Z_t^F, \\ dZ_t^P &= A_P Z_t^P dt + e_P \left(\lambda_F dL_t^F + dL_t^{P,i} \right), & Y_t^P &= b_P^\top Z_t^P. \end{aligned} \quad (17)$$

The parameter λ_F acts only on shocks. It does not change the price drift, the deterministic seasonality, or the marginal CARMA matrices. If the factor driver is represented as $dL_t^F = \sigma_F dW_t^F$ in a Gaussian covariance calculation, the transmitted shock is equivalently $\lambda_F \sigma_F e_P dW_t^F$. The paper uses the dL -driver convention.

The calibration is not performed on the raw deseasonalized residual levels Y^F and Y^P . These correlations are only reported as diagnostics. On hourly intervals, filtered state-space residuals R_i^F and R_i^P are computed from the fixed marginal filters. If dL^F is independent of the idiosyncratic price driver $dL^{P,i}$, their cross-covariance is proportional to λ_F . Define

$$D_{FP} = \frac{1}{n} \sum_{i=1}^n (R_i^F - \bar{R}^F)(R_i^P - \bar{R}^P)^\top$$

as the empirical cross-covariance, and let K_{FP} denote the theoretical cross-covariance generated by the same driver scaling when $\lambda_F = 1$. The scalar estimate is

$$\lambda_F = \frac{b_F^\top D_{FP} b_P}{b_F^\top K_{FP} b_P}. \quad (18)$$

The tables also report the projected state-space residual correlation $\text{Corr}(b_P^\top R^P, b_F^\top R^F)$, which is the empirical correlation associated with the covariance used in eq. (18). The marginal price driver is then decomposed as

$$dL_i^{P,m} = dL_i^{P,i} + \lambda_F dL_i^F, \quad dL_i^{P,i} = dL_i^{P,m} - \lambda_F dL_i^F. \quad (19)$$

After removing $\lambda_F dL^F$, the remaining correlation with the factor driver is used as a direct covariance-level diagnostic for the coupled simulation model.

4.2 Temperature–Price Coupling

The raw correlation between the deseasonalized temperature residual and the deseasonalized German log-price residual is -0.24 . A negative sign is consistent with the winter heating channel, as below-seasonal temperatures tend to increase electricity demand and spot prices. Since the model acts on shocks, this level correlation is only a diagnostic.

For temperature, a positive dL^T is a warmer-than-seasonal shock. A negative λ_T therefore means that warmer-than-seasonal shocks lower price shocks, or equivalently that colder-than-seasonal shocks raise price shocks. This is the expected winter sign, although summer cooling demand can create the opposite local effect. The pooled estimate is

$$\lambda_T = -1.4 \times 10^{-3}.$$

Table 10: Temperature–price coupling diagnostics.

Quantity	Value
$\text{Corr.}(Y^P, Y^T)$	-0.24
$\text{Corr.}(b_P^\top R^P, b_T^\top R^T)$	-0.040
$\text{Corr.}(dL^{P,m}, dL^T)$	-0.042
$\text{Corr.}(dL^{P,i}, dL^T)$	-9.4×10^{-4}
$\text{Std.}(\lambda_T dL^T) / \text{Std.}(dL^{P,i})$	4.1%

After removing $\lambda_T dL^T$, the remaining price driver is close to uncorrelated with the temperature driver, which supports the covariance decomposition used for simulation. The ratio in the table is therefore close to the absolute marginal driver correlation for a simple covariance reason. Let $X = dL^{P,m}$, $Y = dL^{P,i}$, and $Z = \lambda_T dL^T$. Since $X = Y + Z$ and $\text{Cov}(Y, Z) \simeq 0$ after the removal,

$$\text{Cov}(X, Z) = \text{Cov}(Y + Z, Z) \simeq \text{Var}(Z).$$

It follows that $|\text{Corr}(X, Z)| \simeq \text{Std}(Z) / \text{Std}(X)$. Since the transmitted component is small, $\text{Std}(X) \simeq \text{Std}(Y)$, which gives

$$|\text{Corr}(dL^{P,m}, dL^T)| \simeq \frac{\text{Std}(\lambda_T dL^T)}{\text{Std}(dL^{P,i})}.$$

4.3 Solar–Price Coupling

For solar, the raw correlation between the deseasonalized latent residual and the deseasonalized German log-price residual is 0.063 . The solar factor is the deseasonalized clear-sky shortfall variable, so a positive residual corresponds to a larger latent solar shortfall, not to higher solar production.

Because Y^S increases with latent solar shortfall, a positive solar shock represents lower effective solar availability. A positive coupling coefficient would therefore match the merit-order intuition, as lower solar availability raises residual demand and price shocks. Unlike temperature and wind, the raw residual correlation is positive, whereas the projected state-space residual correlation is negative. The coupling estimator is based on the latter covariance, which explains the negative sign of the estimate.

$$\lambda_S = -8.0 \times 10^{-6}.$$

Table 11: Solar–price coupling diagnostics.

Quantity	Value
$\text{Corr.}(Y^P, Y^S)$	0.063
$\text{Corr.}(b_P^\top R^P, b_S^\top R^S)$	−0.0018
$\text{Corr.}(dL^{P,m}, dL^S)$	-1.0×10^{-3}
$\text{Corr.}(dL^{P,i}, dL^S)$	7.8×10^{-4}
$\text{Std.}(\lambda_S dL^S) / \text{Std.}(dL^{P,i})$	0.18%

The solar driver correlation and the transmitted-shock standard-deviation ratio are both close to zero. Solar has the smallest ratio among the three factors, which questions the relevance of a coupled price–solar shock model for this dataset.

4.4 Wind–Price Coupling

For wind, the raw correlation between the logit residual and the German log-price residual is −0.50. The wind transform is monotone in the capacity factor, so a positive wind residual corresponds to above-seasonal wind production. The negative sign is consistent with the merit-order effect.

For wind, a positive dL^W is an above-seasonal production shock. The expected coupling sign is therefore negative, since higher wind production lowers residual demand. The estimate is

$$\lambda_W = -1.7 \times 10^{-3}.$$

Table 12: Wind–price coupling diagnostics.

Quantity	Value
$\text{Corr.}(Y^P, Y^W)$	−0.50
$\text{Corr.}(b_P^\top R^P, b_W^\top R^W)$	−0.040
$\text{Corr.}(dL^{P,m}, dL^W)$	−0.040
$\text{Corr.}(dL^{P,i}, dL^W)$	3.7×10^{-5}
$\text{Std.}(\lambda_W dL^W) / \text{Std.}(dL^{P,i})$	4.0%

The wind channel is the clearest renewable coupling in this calibration. The marginal price driver has a small negative wind correlation, and after removing $\lambda_W dL^W$ the remaining idiosyncratic driver has no detectable linear correlation with wind. The transmitted component remains small, with a standard-deviation ratio close to the temperature one, and this ratio is consistent with the state-space residual correlation.

5 Conclusion

The calibration procedure used in this paper provides accurate continuous-time CARMA marginals for prices, temperature, solar production and wind production. The ACF initialization gives interpretable persistence scales, while the QMLE step improves the likelihood without losing the main autocorrelation structure. Compared with discrete ARMA benchmarks, the CARMA specification keeps a continuous-time state-space representation that is directly usable for joint simulation.

For the coupled model, the relevant dependence is measured at the state-space and driver level rather than on raw residual levels. Temperature and wind transmit small but visible shocks to the price driver, with standard-deviation ratios around 4%, while the solar coupling is much weaker. This questions the relevance of a linear coupled price–solar shock model

on this dataset. More generally, the present framework lets physical factors affect prices only through shocks. Allowing temperature, wind or solar to also enter the price drift could capture slower structural effects, and a multivariate CARMA specification is a natural direction for such an extension.

Appendix

The appendix collects supplementary diagnostics used to audit the empirical workflow. It first reports additional data and deseasonalization checks, then gives the full marginal CARMA calibration outputs, and finally displays the residual scatter plots used to assess the coupling step.

A Data and Deseasonalization Diagnostics

This appendix reports additional preprocessing diagnostics that are not shown in the main text. Time-domain plots use the common window from 1 June 2025 to 8 June 2025. The price daily factor is shown over the first semester of 2025, because it is estimated at daily frequency.

A.1 Electricity Prices

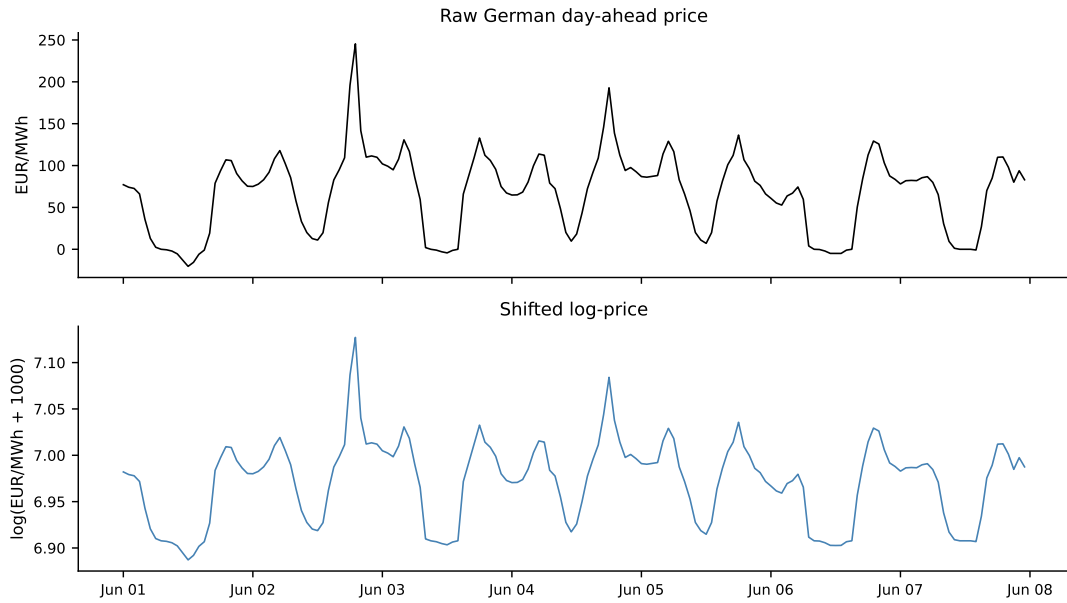


Figure A.1: German day-ahead price and shifted log-price over the common appendix week. The shift makes the logarithmic coordinate well-defined when hourly prices are negative.

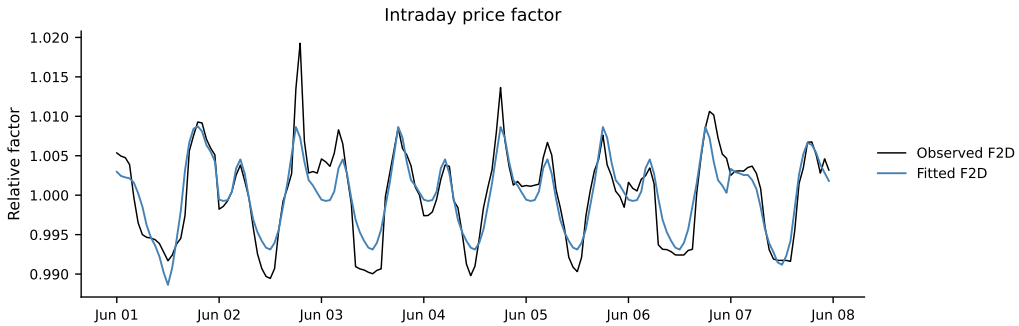


Figure A.2: Observed and fitted intraday price factor over the common appendix week. The observed factor is the shifted log-price divided by its daily mean.

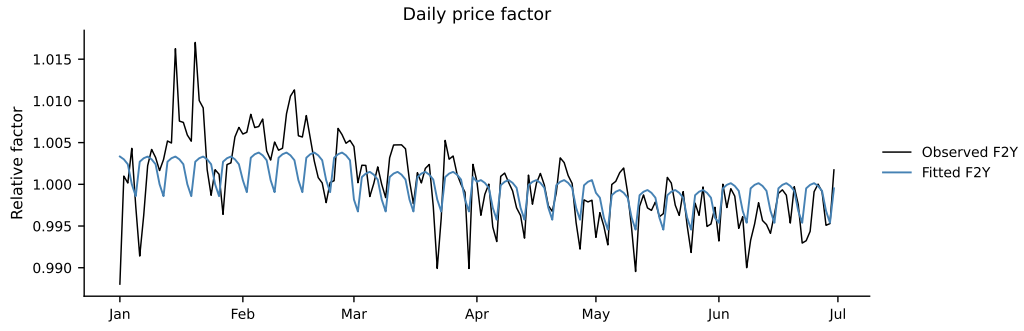


Figure A.3: Observed and fitted daily price factor over the first semester of 2025. The observed factor is the daily mean shifted log-price divided by its yearly mean.

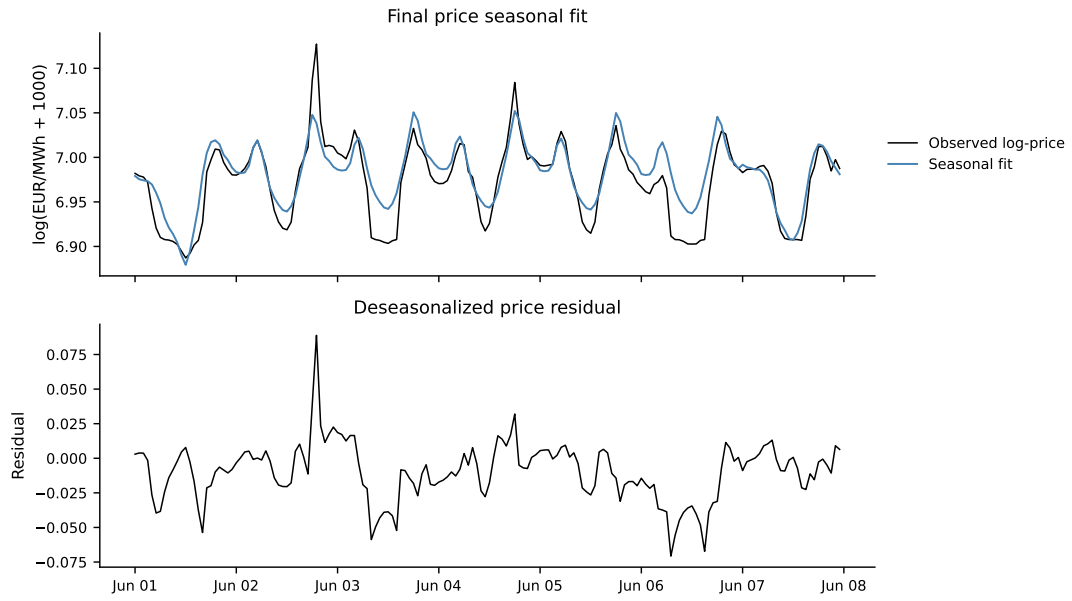


Figure A.4: Final deterministic price seasonality and resulting deseasonalized log-price residual over the common appendix week.

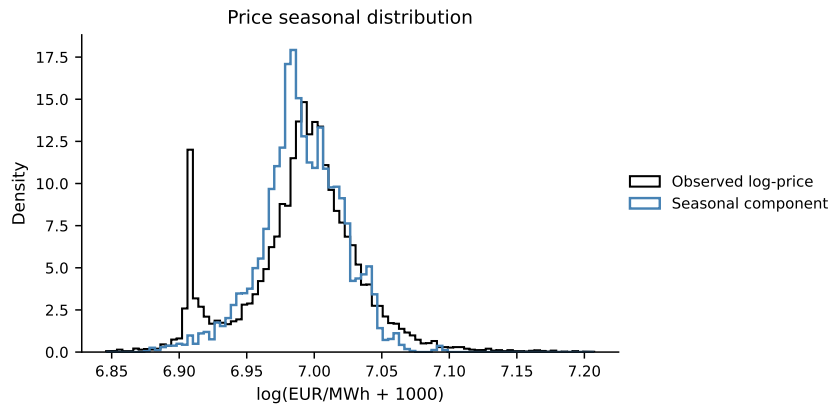


Figure A.5: Marginal distribution of the shifted log-price and of its fitted deterministic seasonal component over the 2023–2025 calibration sample.

A.2 Temperature

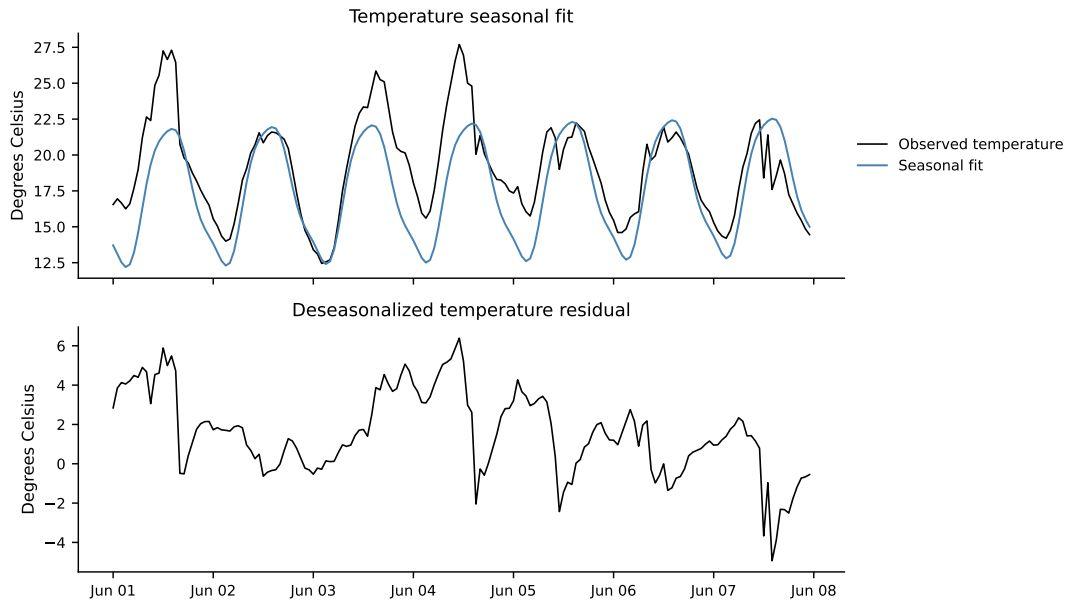


Figure A.6: Temperature seasonal fit and resulting deseasonalized residual over the common appendix week.

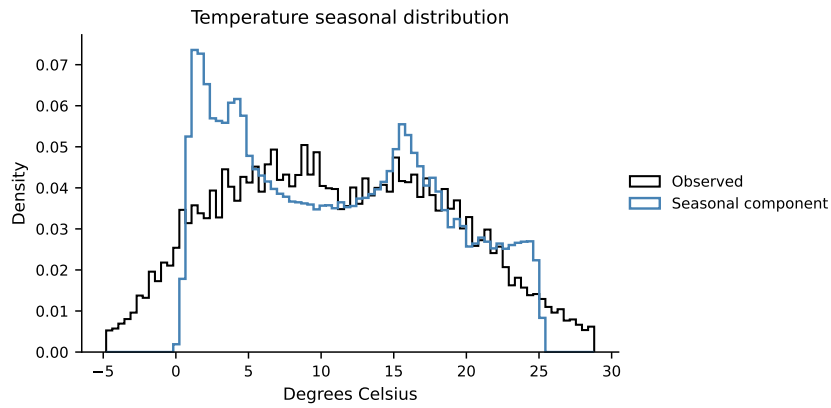


Figure A.7: Marginal distribution of observed temperature and of the fitted deterministic seasonal component over the 2023–2025 calibration sample.

A.3 Solar Capacity Factors

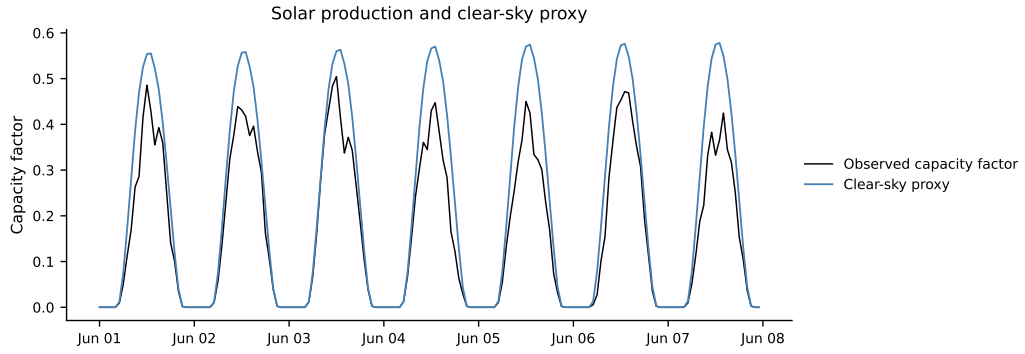


Figure A.8: Observed solar capacity factor and clear-sky proxy over the common appendix week. The clear-sky proxy is the deterministic envelope used before constructing the latent logit coordinate.

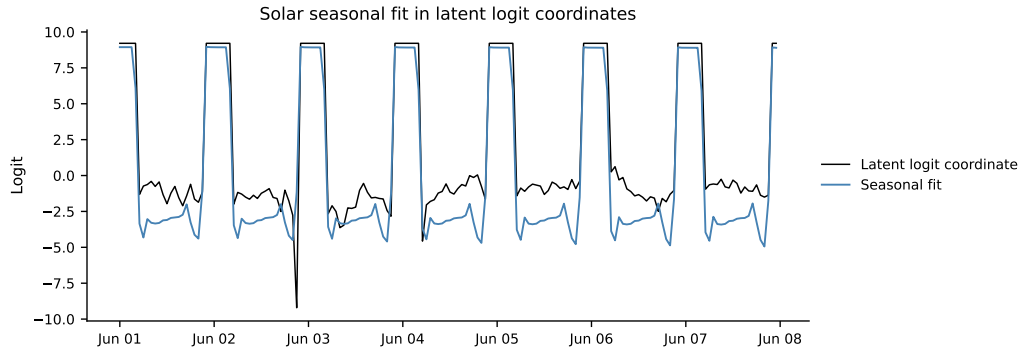


Figure A.9: Solar seasonal fit in the latent logit coordinate over the common appendix week. This is the coordinate in which the deterministic seasonality is estimated.

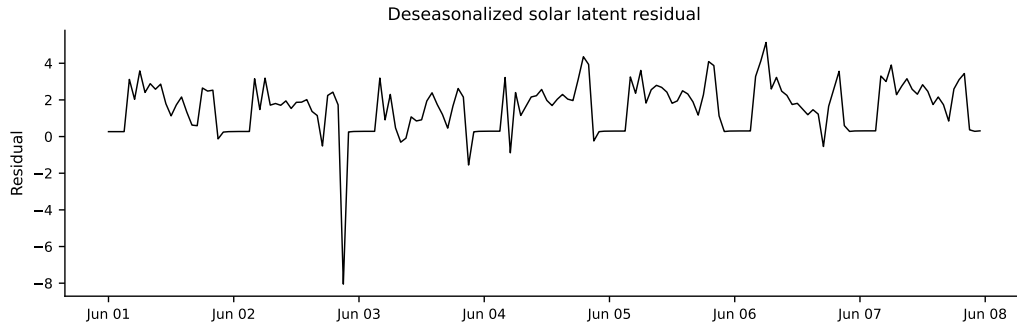


Figure A.10: Deseasonalized solar residual in the latent logit coordinate over the common appendix week. This residual is the stochastic component calibrated with a CARMA model.

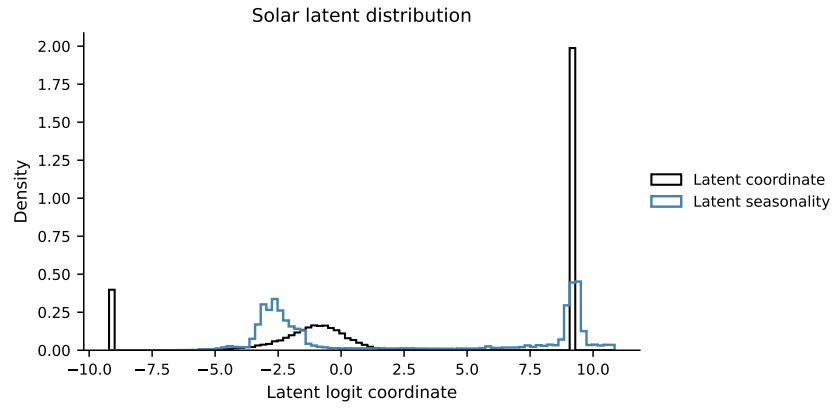


Figure A.11: Marginal distribution of the solar latent coordinate and of its fitted deterministic seasonal component over the calibration sample.

A.4 Wind Capacity Factors

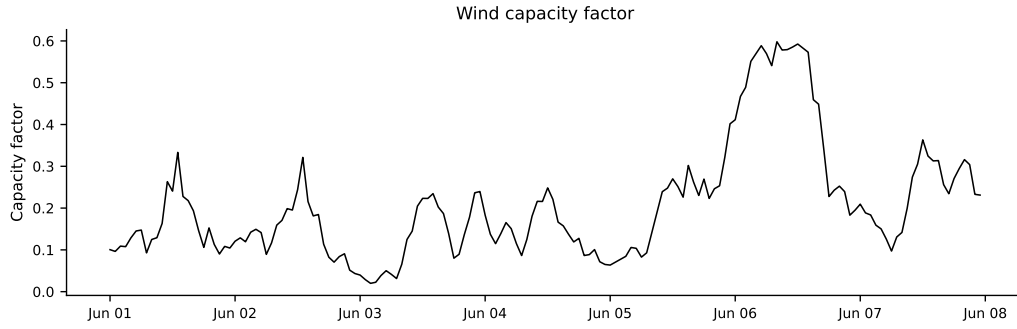


Figure A.12: Observed wind capacity factor over the common appendix week.

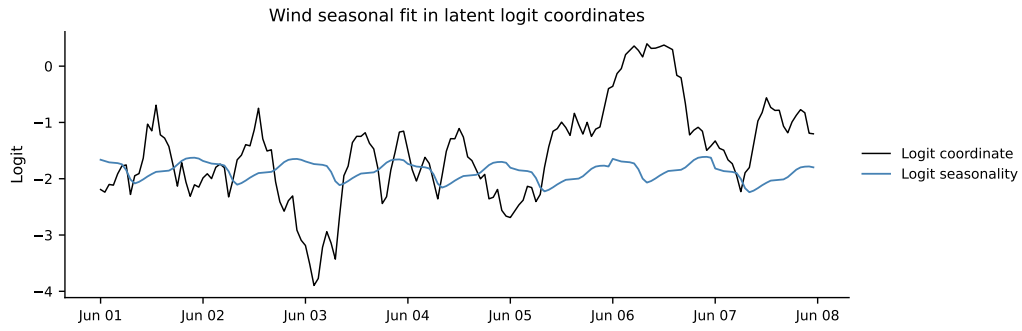


Figure A.13: Wind seasonal fit in the latent logit coordinate over the common appendix week. This is the coordinate in which the deterministic seasonality is estimated.

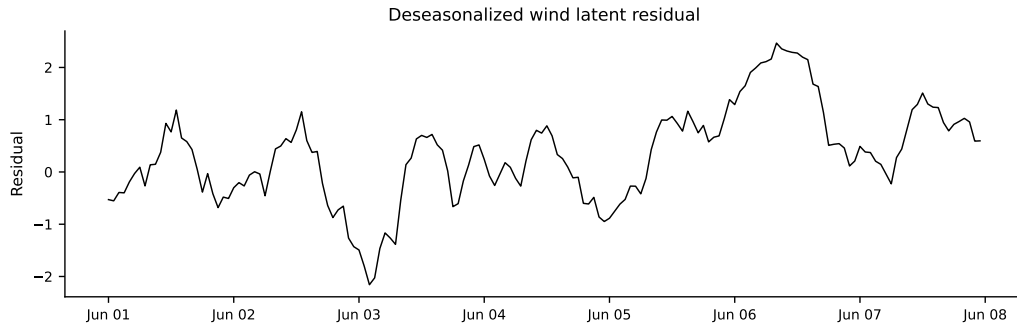


Figure A.14: Deseasonalized wind residual in the latent logit coordinate over the common appendix week. This residual is the stochastic component calibrated with a CARMA model.

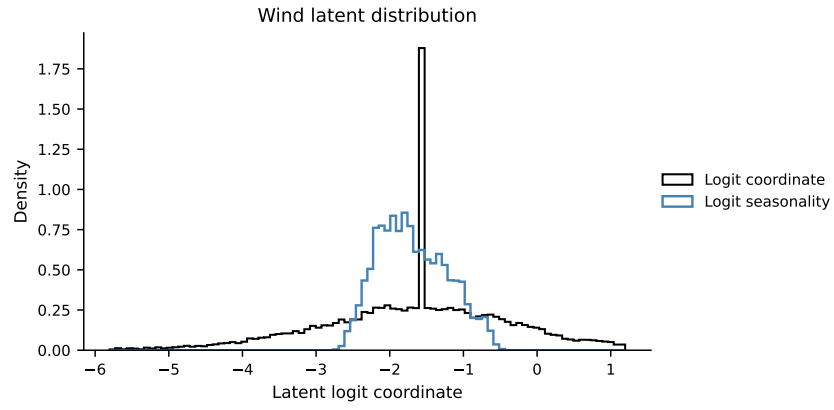


Figure A.15: Marginal distribution of the wind latent logit coordinate and of its fitted deterministic seasonal component over the calibration sample.

B Marginal Calibration Diagnostics

B.1 German Spot Prices

Table B.1: Full German spot price CARMA(5,4) calibration parameters.

Quantity	ACF initialization	QMLE
Log-likelihood	77562	77709
Levy drift m	1.40×10^{-5}	8.47×10^{-6}
Levy variance ν^2	1.88×10^{-4}	1.75×10^{-4}
λ_1	-0.263	-0.153
λ_2	-0.0242	-0.0193
λ_3	-2.41×10^{-4}	-3.51×10^{-4}
λ_o	$-0.00400 \pm 0.262i$	$-0.00400 \pm 0.262i$
a_1	0.296	0.181
a_2	0.0773	0.0730
a_3	0.0198	0.0119
a_4	4.41×10^{-4}	2.07×10^{-4}
a_5	1.05×10^{-7}	7.11×10^{-8}
b_0	7.84×10^{-6}	4.89×10^{-6}
b_1	0.00719	0.00387
b_2	0.0713	0.0672
b_3	0.130	0.0809
b_4	1.00	1.00

Table B.2: NIG fit for recovered hourly Levy increments in the German spot price model.

Quantity	Value
α	47.5
β	-5.07
δ	0.00687
μ	7.40×10^{-4}
γ	47.2
Mean	1.89×10^{-6}
Standard deviation	0.0121
Skewness	-0.563
Excess kurtosis	9.68
Log-likelihood	83442

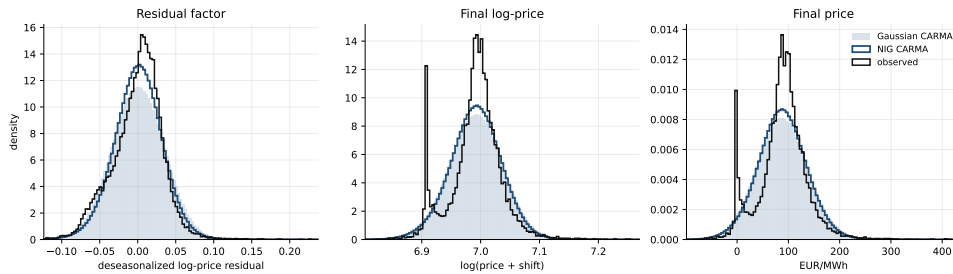


Figure B.1: Distribution checks for the German price marginal model at the residual factor, shifted log-price, and EUR/MWh price levels.

B.2 Temperature

Table B.3: Full temperature CARMA(4,3) calibration parameters.

Quantity	ACF initialization	QMLE
Log-likelihood	−23760	−22941
Levy drift m	5.48×10^{-4}	2.85×10^{-4}
Levy variance ν^2	0.350	0.161
λ_1	−0.0272	−1.31
λ_2	−0.0119	−0.0182
λ_o	$-0.00953 \pm 0.262i$	$-0.00953 \pm 0.262i$
a_1	0.0582	1.35
a_2	0.0697	0.118
a_3	0.00269	0.0915
a_4	2.23×10^{-5}	0.00163
b_0	0.00136	0.163
b_1	0.0690	0.201
b_2	0.0782	2.57
b_3	1.00	1.00

Table B.4: NIG fit for recovered hourly Levy increments in the temperature model.

Quantity	Value
α	2.06
β	−0.257
δ	0.299
μ	0.0372
γ	2.05
Mean	-2.58×10^{-4}
Standard deviation	0.385
Skewness	−0.477
Excess kurtosis	5.20
Log-likelihood	−9716

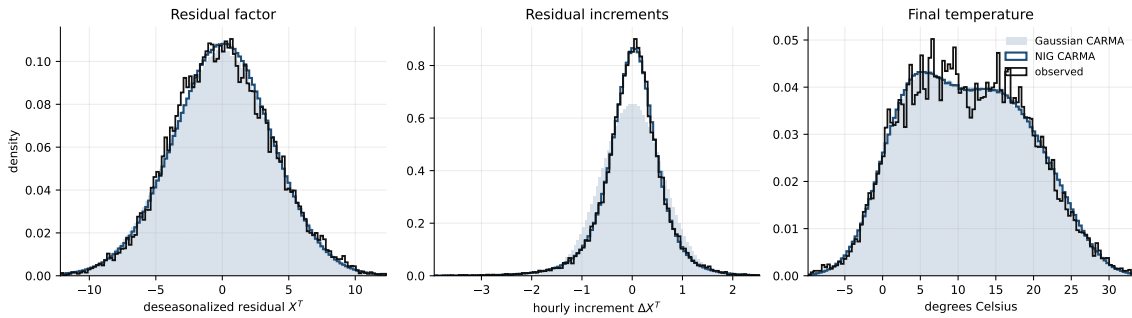


Figure B.2: Distribution checks for the German temperature marginal model at the deseasonalized residual, hourly residual-increment, and physical temperature levels.

B.3 Solar Capacity Factor

Table B.5: Full solar capacity factor CARMA(4,3) calibration parameters.

Quantity	ACF initialization	QMLE
Log-likelihood	−84637	−84623
Levy drift m	1.14×10^{-4}	1.27×10^{-4}
Levy variance ν^2	9.27	9.12
λ_1	−1.37	−1.35
λ_2	−0.0166	−0.0184
λ_o	$-0.00984 \pm 0.262i$	$-0.00984 \pm 0.262i$
a_1	1.41	1.39
a_2	0.119	0.121
a_3	0.0957	0.0947
a_4	0.00156	0.00171
b_0	0.00848	0.00899
b_1	0.0836	0.0819
b_2	0.231	0.239
b_3	1.00	1.00

Table B.6: NIG fit for recovered hourly Levy increments in the solar capacity factor model.

Quantity	Value
α	0.136
β	−0.0149
δ	1.24
μ	0.136
γ	0.135
Mean	1.33×10^{-4}
Standard deviation	3.04
Skewness	−0.803
Excess kurtosis	18.8
Log-likelihood	−91910

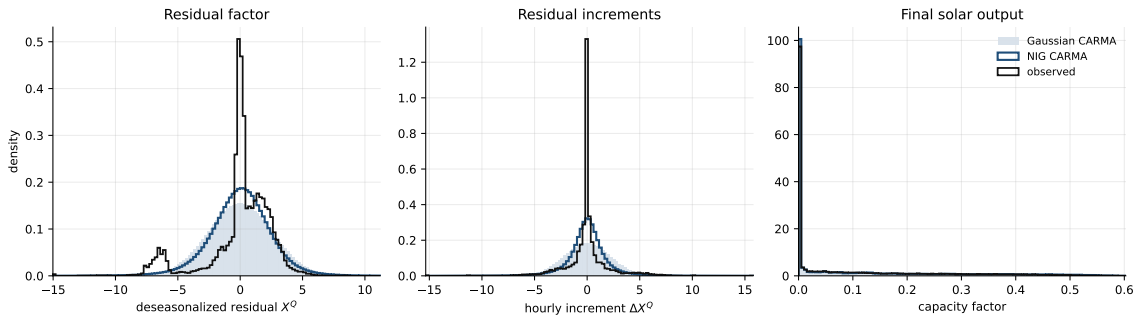


Figure B.3: Distribution checks for the German solar marginal model at the deseasonalized latent residual, hourly residual-increment, and physical capacity-factor levels.

B.4 Wind Capacity Factor

Table B.7: Full wind capacity factor CARMA(4,3) calibration parameters.

Quantity	ACF initialization	QMLE
Log-likelihood	−7013	−6984
Levy drift m	6.36×10^{-5}	1.57×10^{-5}
Levy variance ν^2	0.102	0.0993
λ_1	−0.0431	−0.0331
λ_2	−0.00388	-7.68×10^{-4}
λ_o	$-0.00880 \pm 0.262i$	$-0.00880 \pm 0.262i$
a_1	0.0646	0.0515
a_2	0.0696	0.0692
a_3	0.00323	0.00233
a_4	1.15×10^{-5}	1.75×10^{-6}
b_0	4.27×10^{-4}	9.23×10^{-5}
b_1	0.0685	0.0689
b_2	0.0472	0.0539
b_3	1.00	1.00

Table B.8: NIG fit for recovered hourly Levy increments in the wind capacity factor model.

Quantity	Value
α	1.53
β	−0.0851
δ	0.132
μ	0.00707
γ	1.53
Mean	-2.78×10^{-4}
Standard deviation	0.294
Skewness	−0.370
Excess kurtosis	15.0
Log-likelihood	1625

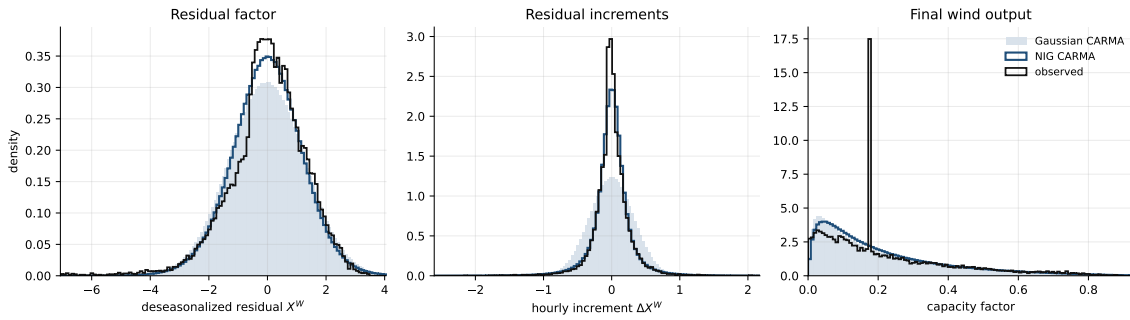


Figure B.4: Distribution checks for the German wind marginal model at the deseasonalized logit residual, hourly residual-increment, and physical capacity-factor levels.

C Coupling Diagnostics

The figures in this appendix compare the raw deseasonalized residual levels with the projected state-space residuals used by the covariance estimator in section 4. The raw plots describe visible marginal co-movement, whereas the projected plots correspond to the CARMA state residuals on which the coupling coefficient is estimated.

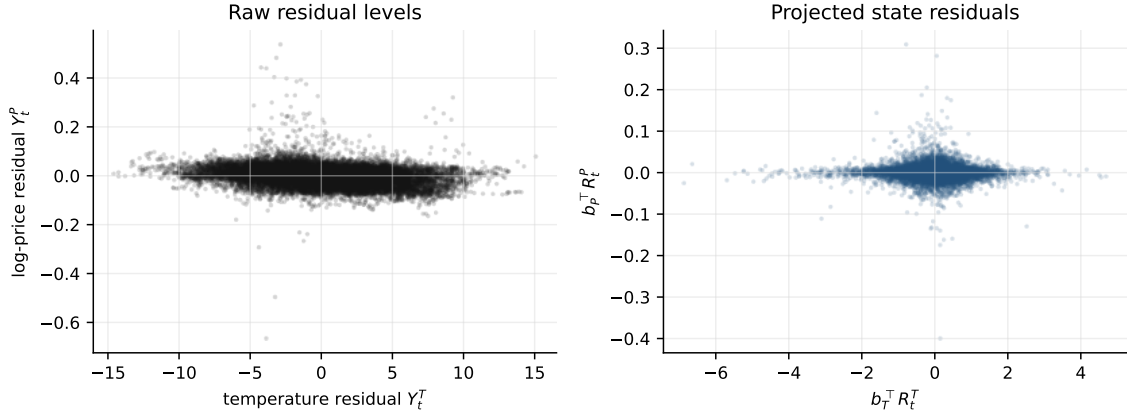


Figure C.1: Temperature–price residual scatter plots. The first panel uses raw deseasonalized residual levels. The second panel uses the projected state-space residuals entering the covariance-based coupling estimator.

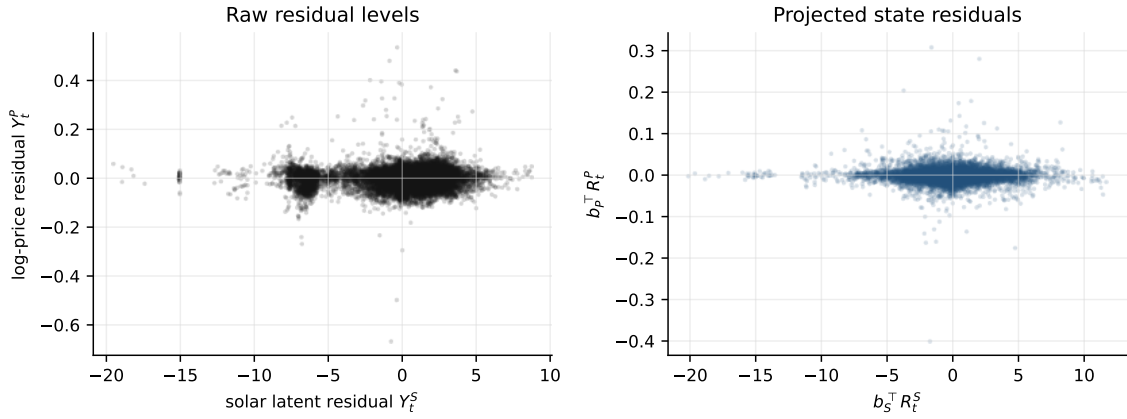


Figure C.2: Solar–price residual scatter plots. The raw residual is the solar latent shortfall coordinate, while the projected residual is obtained from the calibrated CARMA state-space filter.

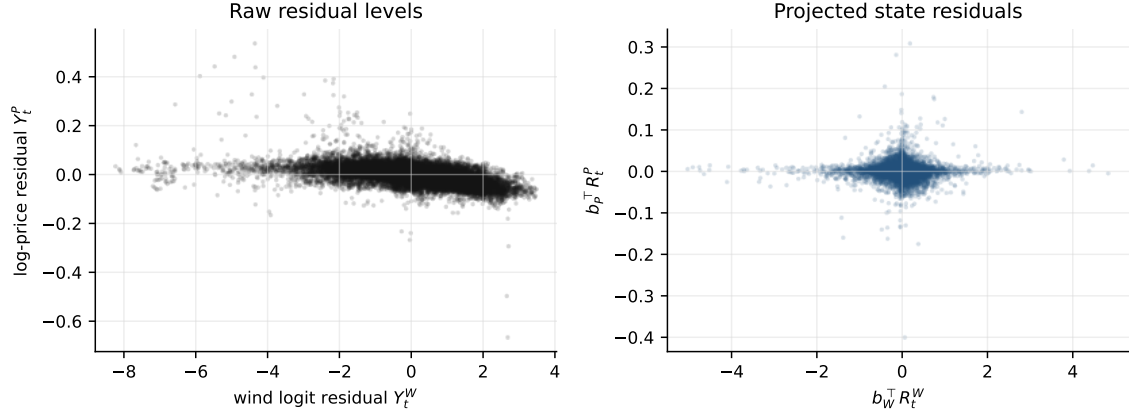


Figure C.3: Wind–price residual scatter plots. The first panel uses wind logit residual levels and the second panel uses the corresponding projected CARMA state-space residuals.

References

- [1] Florentina Paraschiv. Price dynamics in electricity markets. In *Handbook of Risk Management in Energy Production and Trading*, pages 47–69. Springer US, Boston, MA, 2013. ISBN 9781461490357. doi: 10.1007/978-1-4614-9035-7_3.
- [2] Peter J. Brockwell and Alexander M. Lindner. Lévy-driven CARMA processes: Definition, existence, uniqueness and properties. In *Continuous-Parameter Time Series*, pages 410–442. De Gruyter, 2024. ISBN 9783111325033. doi: 10.1515/9783111325033-018.
- [3] Peter J. Brockwell and Alexander M. Lindner. QML estimation for CARMA processes. In *Continuous-Parameter Time Series*, pages 443–478. De Gruyter, 2024. ISBN 9783111325033. doi: 10.1515/9783111325033-019.